Supplemental Material

Quantum-Geometric Bound on Dynamical Instability in Bosonic Systems

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This document collects the derivations, numerical protocols and supporting calculations referred to in the main text. Equation, figure and table numbers prefixed by a letter belong to this document; unprefixed numbers refer to the main text. References are those of the reference list of the main text.

S1. Closed-system dynamics along Family A, extended range

Table I of the main text gives both reciprocal families over the range shown in Fig. 1. Table S1 continues Family A beyond that range, to make explicit that the exponent stays fixed while the short-time fidelity decay grows without bound within the quadratic model: over the points listed the asymptotic growth rate is 2.000000 in units of λ_0 at every point, reproduced to six digits, while the fidelity decay spans two orders of magnitude.

TABLE S1. Family A continued. Rates and times in units of λ_0; the fidelity decay is evaluated at $\lambda_0 t = 0.02$, well inside the quadratic regime. The first four rows are those of Table I of the main text.

δ/λ_0	$\sqrt{2}\, v_{FS}/\lambda_0$	$1 - F$	$\mathrm{d}\ln\langle n\rangle/\mathrm{d}t$
0	1.000	2.000×10^{-4}	2.000000
1	1.414	3.998×10^{-4}	2.000000
2	2.236	9.986×10^{-4}	2.000000
3	3.162	1.994×10^{-3}	2.000000
5	5.099	5.160×10^{-3}	2.000000
10	10.050	1.961×10^{-2}	2.000000

Along Family A the members also differ in the shape of the covariance, with Var x = ½[$\cosh^2\lambda t$ + $((\delta - \varepsilon)^2/\lambda^2)\sinh^2\lambda t$] and Var p the same expression with $\delta + \varepsilon$, so that at $\lambda_0 t = 0.8$ the quadrature anisotropy across the four members of Table I of the main text runs 24.5, 67.0, 313.9 and 1123.6 while the squeezing axis rotates from 135.0° to 103.3°.

S2. The Gaussian variance formula

For a quadratic generator $\hat{H}$ = ½ $\hat{z}^{\mathrm{T}} G \hat{z}$ and a Gaussian state of covariance matrix Σ with vanishing first moments, Wick's theorem for the fourth moments of the quadratures gives

$$\mathrm{Var}(\hat{H}) = \tfrac{1}{2}\,\mathrm{Tr}(G\Sigma G\Sigma) + \tfrac{1}{8}\,\mathrm{Tr}(GJGJ), \quad \text{(S1)}$$

the second term arising from the commutator contribution to the symmetrized product. This is the standard Gaussian expression [21,22] and is the only input to Theorem 1 beyond the Anandan–Aharonov relation.

Equation (S1) was validated independently against a closed form: for a squeezed vacuum with $\langle \hat{n} \rangle = \sinh^2 r$, evaluating Eq. (S1) for the number operator reproduces $2\,\langle \hat{n} \rangle\,(\langle \hat{n} \rangle + 1)$ to 10^{-10} across the range of r tested. It was further checked against direct Fock-basis evaluation for the generator of Eq. (4): with the mode truncated at 400 levels, the exact bare-vacuum variance equals $\varepsilon^2/2$ to twelve digits at δ = 0, 0.7, 0.99 and 2.0 in units of ε, confirming the detuning independence of Eq. (5) outside the symplectic formalism used to derive it.

One numerical point should be noted, since the closed form is the reliable route. Because the generator of Eq. (4) is time independent, Var($\hat{H}$) is exactly conserved along the flow. Propagating the covariance matrix to large λt and then taking the trace is ill-conditioned, the result being a difference of exponentially large and exponentially small entries; the closed form should be used, with any propagated evaluation restricted to moderate λt.

S3. Numerical verification

S3.1 Theorems 1 and 2

Theorem 1 is an algebraic identity and admits no numerical residual beyond floating-point rounding, which tests an implementation rather than the statement. For each N from 1 to 6, five hundred real symmetric matrices G of dimension $2N$ were drawn with independent normal entries and symmetrized, the left-hand side of Eq. (2) was evaluated through Eq. (S1) at $\Sigma = \tfrac{1}{2}\cdot 1$, and the right-hand side through $\mathrm{Tr}\, G^2$ and $\mathrm{Tr}(JG)^2$. The maximum absolute discrepancy is exactly zero in double precision at every N, the two expressions reducing to the same sequence of floating-point operations.

Corollary 2 was checked directly: for $N = 1$, 2 and 3, symmetric generators G_p commuting with J were constructed by projection and added to random base generators with coefficients up to 12. The symplectic condition $M^\mathrm{T}J + JM = 0$ holds throughout to 10^{-10}, v_{FS} is unchanged to all displayed digits at every coefficient, and the largest exponent varies over the full range, reaching zero. Theorem 2 was verified step by step as an implementation check rather than as evidence, the statement being proved. The anticommutation $SJ + JS = 0$ holds to machine zero for $N = 1$ to 6; the spectrum of S pairs as $\pm\mu$ to 1.7×10^{-14}; the inequality $\|S\|_\mathrm{F}^2 \geq 2\mu_{\max}^2(S)$ is satisfied with residual no worse than -7×10^{-15}; and over 1.2×10^5 random generators at $N = 1$ to 6 neither the Bendixson step nor Eq. (3) is violated, the largest sampled ratio falling short of unity by 3×10^{-9}, an extremal order statistic of the sample rather than a statement of precision. Random Gaussian generators do not approach saturation for $N > 1$, whereas the structured saturating family of the proof — one resonant squeezer with spectator modes — returns unity at every N from 1 to 5, with a residual of one unit in the last place, 2×10^{-16}.

The closed form of Eq. (7) was checked against direct propagation of the covariance matrix under Eq. (4) at representative points of both families, with relative agreement better than 10^{-13} over the range $0 \leq \lambda t \leq 8$; the quadrature expressions quoted in the main text agree with the propagated covariance to the digits shown.

For the single-mode generator of Eq. (4) the identity may be verified by hand: $G = \mathrm{diag}(\delta + \varepsilon, \delta - \varepsilon)$ gives $\mathrm{Tr}\, G^2 = 2\delta^2 + 2\varepsilon^2$; $M = JG$ has $M^2 = -(\delta^2 - \varepsilon^2)\cdot 1$ and hence $\mathrm{Tr}\, M^2 = 2(\varepsilon^2 - \delta^2)$; the sum is $4\varepsilon^2$ and Eq. (2) returns $v_{FS} = |\varepsilon|/\sqrt{2}$.

S3.2 Isospectral families and the generic spread

Corollary 1 is established by the following construction at $N = 2$, which varies the generator relative to a fixed bare-vacuum reference. A passive symplectic change of representation, applied simultaneously to generator and state, leaves v_{FS} invariant and is not in question. The construction here is active in the sense of the main text: the bare-vacuum covariance is held fixed while the generator is replaced by a symplectically related one, producing physically distinct generator–state pairs sharing an identical flow spectrum. Along a one-parameter family of such replacements the largest exponent is unchanged by construction, with numerical variation below 2.2×10^{-16}, while v_{FS} rises by a factor 11.1 over the range shown, and without bound within the quadratic model [Fig. S1(a)].

Equivalently, moving the inverse symplectic transformation from the generator onto the state, the same algebra may be read as one fixed generator probed on a family of squeezed Gaussian references, in which v_{FS} is sensitive to the fluctuations of the state it is referred to whereas the exponents of the linear flow are not. This is an equivalent representation of the same construction; the convention used in the main text holds the reference covariance fixed and transforms the generator. That is the content of the construction rather than an artifact of it. Drawing generators at random corroborates the conclusion without depending on any ensemble: among samples whose geometric speed agrees to within 2 %, the largest exponent covers essentially the whole interval permitted by Theorem 2, from values indistinguishable from zero at numerical precision up to 0.99 of the ceiling [Fig. S1(b)]. The coverage is quoted rather than a ratio of largest to smallest exponent within the band, because that ratio is fixed by the single smallest draw and is not a stable statistic.

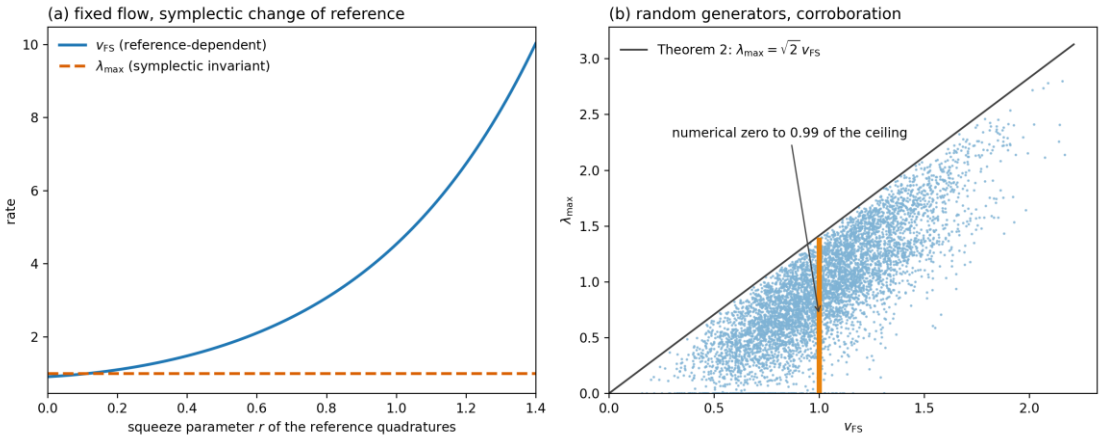


FIG. S1. The geometric speed is not a function of the instability spectrum. (a) A two-mode quadratic generator is transformed through a one-parameter family of active symplectic squeezes while the bare-vacuum covariance is held fixed. The resulting flow matrices are similar to one another and therefore share an identical spectrum, so the family consists of physically distinct generator–state pairs rather than of one system in different coordinates, while v_{FS} changes. Dashed: the largest exponent $\lambda_{\max}$, constant by construction, with numerical variation below 2.2×10^{-16}. Solid: the Fubini–Study speed referred to the bare-mode vacuum of the transformed quadratures, rising by a factor 11.1 over the range shown. (b) Corroboration without the construction: 6000 random two-mode generators, with the exact single-mode on-resonance relation $\lambda = \sqrt{2}\, v_{FS}$ shown as a line that bounds the cloud rather than describing it; the heavy bar spans the exponents found among samples whose speed lies within 2 % of unity, from numerical zero to 0.99 of the ceiling.

S4. Chirped mode — frozen exponent against squeezing rate

For a single mode of instantaneous frequency $\Omega(t)$ the self-referencing non-adiabaticity parameter

$$\eta(t) = |\mathrm{d}\Omega/\mathrm{d}t|/\Omega^2(t) \quad \text{(S2)}$$

is dimensionless by construction and, with the driven-mode conventions of Refs. [30,31], which are preprints and not peer reviewed, the Fubini–Study metric of the instantaneous vacuum gives $g_{tt} = \eta^2\Omega^2/8$ and hence $v_{FS} = \eta\Omega/\sqrt{8}$. Linearizing the driven dynamics gives $\delta\dot{a} = -\mathrm{i}\Omega\delta a + B(t)\delta a^*$ with $|B| = |\mathrm{d}\Omega/\mathrm{d}t|/2\Omega$, generated by $\hat{H} = \hat{H}_{\parallel} + \hat{H}_{\perp}$ with

$\hat{H}_{\parallel} = \hbar\Omega\, \hat{a}^{\dagger}\hat{a}$, $\hat{H}_{\perp} = (\mathrm{i}\hbar/2)(B\, \hat{a}^{\dagger 2} - B^{*}\hat{a}^{2})$. (S3)

In the instantaneous vacuum $\hat{H}_{\parallel}$ generates no motion in projective Hilbert space, since it stabilizes phase-space rotations, and all geometric motion comes from $\hat{H}_{\perp}$, whose energy uncertainty there is $\Delta\hat{H}_{\perp} = \hbar|B|/\sqrt{2}$. Accordingly

$b(t) \equiv (\mathrm{d}\Omega/\mathrm{d}t)/2\Omega$, $\kappa_s \equiv |b| = |B| = \sqrt{2}\, v_{FS} = \eta\Omega/2$ (S4)

is the squeezing rate of the non-adiabatic component, the subscript distinguishing it from the resonator linewidth κ of the Appendix to the main text; the relation $\kappa_s = \sqrt{2}\, v_{FS}$ is an exact algebraic identity requiring no numerical confirmation. The phase convention is chosen so that $B(t) = b(t)$ is real; it is not restricted to be positive, its modulus being $|B| = \kappa_s$ while its sign follows that of $\mathrm{d}\Omega/\mathrm{d}t$. A nonzero phase would rotate the squeezing axis without changing $|B|$. The signed coefficient is kept separate from its modulus because the frozen eigenvalues depend only on b^2 while the time-ordered evolution depends on the sign, which alternates whenever the drive is not monotonic.

Dropping $\hat{H}_{\parallel}$ is legitimate for the Fubini–Study speed of the instantaneous vacuum, because that term is a stabilizer there. It is not legitimate for the character of the flow. In the quadratures $(\hat{x}, \hat{p})$ the effective generator produces the traceless flow matrix $M = [[b, \Omega], [-\Omega, -b]]$, with instantaneous eigenvalues $\pm\lambda_{\mathrm{loc}}$,

$\lambda_{\mathrm{loc}}^2 = b^2 - \Omega^2 = \kappa_s^2 - \Omega^2 = \Omega^2(\eta^2/4 - 1)$. (S5)

The frozen generator belongs to the elliptic conjugacy class of $\mathrm{SL}(2, \mathbb{R})$ for $\eta < 2$ and to the hyperbolic class for $\eta > 2$; the hyperbolic character of $\hat{H}_{\perp}$ taken alone is an artifact of having removed $\hat{H}_{\parallel}$. Equivalently, the quadratic form has eigenvalues $\Omega \pm \kappa_s$, the smaller changing sign exactly at $\eta = 2$. This concerns the effective generator governing the Bogoliubov amplitudes in the instantaneous-mode frame, in which $\hat{H}_{\perp}$ arises from the frame transformation itself; the underlying physical oscillator, with Hamiltonian $\hat{p}^2/2 + \Omega^2(t)\hat{x}^2/2$, remains positive definite and retains a well-defined instantaneous ground state whenever $\Omega^2(t) > 0$, so $v_{FS} = \eta\Omega/\sqrt{8}$ does not cease to exist when the effective flow enters the local hyperbolic class. In a statically pumped realization such as Eq. (4), where the squeezing term belongs to the physical

Hamiltonian rather than to a frame transformation, the indefiniteness at the same threshold is a physical statement.

Table S2 quantifies the systematic overestimate incurred whenever κ_s is substituted for the local hyperbolic rate. At $\eta = \sqrt{8}$, where the geometric speed equals the spectral scale, the squeezing rate exceeds λ_{loc} by $\sqrt{2}$: stated relative to λ_{loc} the overestimate is 41.4 %, stated relative to κ_s it is 29.3 %. The two conventions differ and should not be mixed; the first is quoted throughout. The approach to agreement is slow.

TABLE S2. Squeezing rate of the non-adiabatic component against the instantaneous frozen-flow exponent of the effective quadratic generator, for the chirped-mode parametrization of Eq. (S2). No real λ_{loc} exists below $\eta = 2$. The last column is the overestimate relative to λ_{loc}.

η	κ_s/Ω	$\lambda_{\mathrm{loc}}/\Omega$	$\kappa_s/\lambda_{\mathrm{loc}}$	$(\kappa_s - \lambda_{\mathrm{loc}})/\lambda_{\mathrm{loc}}$
2.5	1.250	0.750	1.667	66.7 %
$\sqrt{8} = 2.828$	1.414	1.000	1.414	41.4 %
3	1.500	1.118	1.342	34.2 %
4	2.000	1.732	1.155	15.5 %
10	5.000	4.899	1.021	2.1 %

Figure S2 shows the two rates against the non-adiabaticity, together with the inverse of the ratio of Eq. (6) for the autonomous counterpart of the same algebra.

The symplectic normal form of the frozen flow was checked separately. In the hyperbolic regime $\kappa_s > \Omega$ the flow matrix lies in the hyperbolic conjugacy class of SL(2, $\mathbb{R}$) and is conjugate, by a real symplectic transformation, to $\lambda_{\mathrm{loc}}\,\mathrm{diag}(1, -1)$. Constructing the eigenvector matrix V and normalizing so that $V^{\mathrm{T}}JV = J$, direct evaluation at $\eta = 2.5$, 4 and 10 gives $V^{\mathrm{T}}JV = J$ to within 1.1×10^{-16} and $V^{-1}MV = \lambda_{\mathrm{loc}}\,\mathrm{diag}(1, -1)$ to within 8.9×10^{-16}, with $\lambda_{\mathrm{loc}} = 0.7500$, 1.7321 and 4.8990 in units of Ω respectively. The frozen effective generator above threshold is therefore canonically a dilatation, but of rate λ_{loc} and not of rate κ_s.

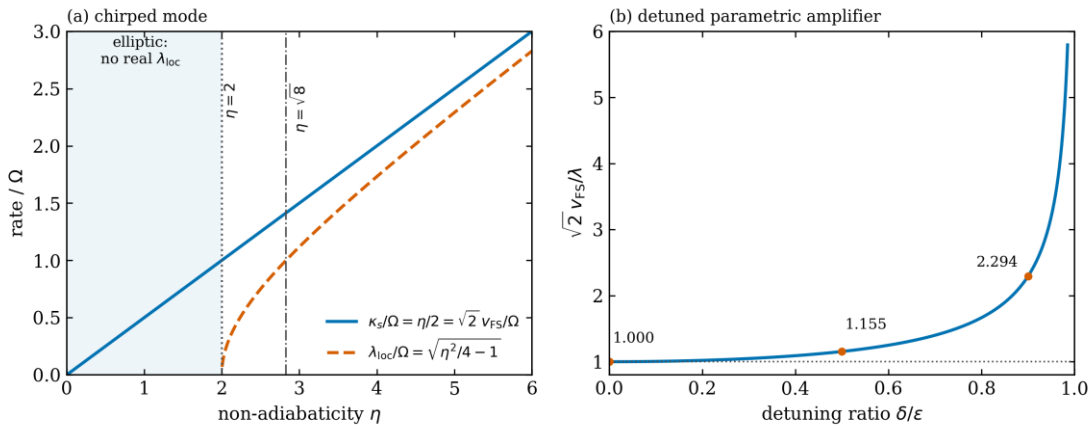


FIG. S2. Chirped mode and its autonomous counterpart. (a) Solid, the squeezing rate $\kappa_s/\Omega = \eta/2 = \sqrt{2}\, v_{FS}/\Omega$, defined for all η; dashed, the frozen-flow exponent $\lambda_{loc}/\Omega = \sqrt{(\eta^2/4 - 1)}$ of Eq. (S5), real only in the unshaded region $\eta > 2$. The dotted vertical line marks $\eta = 2$, where the effective quadratic generator becomes indefinite; the dash-dotted line marks $\eta = \sqrt{8}$, where the geometric speed equals the spectral scale. The gap between the curves is the overestimate of Table S2 and closes only as $\eta \to \infty$. (b) The inverse of the ratio of Eq. (6) for the detuned amplifier, $\sqrt{2}\, v_{FS}/\lambda = [1 - \delta^2/\varepsilon^2]^{-1/2}$, equal to unity on resonance and diverging at threshold; this is the quotient of the geometric-speed curve by the instability-rate curve of Fig. 1(b), shown here for comparison with panel (a) rather than as an independent result.

S5. Failure of locality at parametric resonance

For a genuinely time-dependent chirp the generator is non-autonomous and the true growth of the tangent vector is governed by the time-ordered propagator, not by the spectrum of M at any single instant. It is a standard fact for non-autonomous linear systems that the eigenvalues of the frozen matrix do not in general determine finite-time or asymptotic Lyapunov exponents, and can even fail to determine stability. Accordingly λ_{loc} of Sec. S4 is called the instantaneous frozen-flow exponent and is not identified with a Lyapunov exponent of the chirped dynamics; it becomes a genuine exponential rate only when the coefficients are constant, as in the main text.

The gap can be total rather than quantitative. Take a modulated frequency at parametric resonance, $\Omega(t) = \Omega_0[1 + m\cos(2\Omega_0 t)]$, and integrate the effective generator of Sec. S4 with the

signed coefficient $b(t)$. Since the modulation is periodic with period $T_p = \pi/\Omega_0$, the growth is characterized exactly by the largest Floquet exponent $\mu_F = T_p^{-1} \ln \rho[\Phi(T_p, 0)]$, with ρ the spectral radius of the monodromy matrix, independent of any choice of norm and of the integration time. The integration used relative tolerance 10^{-12}; the monodromy satisfies det $\Phi(T_p, 0) = 1$ to 10^{-12}, as a symplectic propagator must. Table S3 collects the comparison, with rates in units of Ω_0.

TABLE S3. Frozen-flow prediction against the exact Floquet exponent for the modulated frequency $\Omega(t) = \Omega_0[1 + m \cos(2\Omega_0 t)]$.

m	max η	frozen prediction	μ_F	$T^{-1} \ln \|\|\Phi\|\|$
0.02	0.040	0	0.009999	0.01000
0.05	0.101	0	0.024977	0.02499
0.10	0.204	0	0.049817	0.04984
0.20	0.433	0	0.098550	0.09860
0.40	1.111	0	0.188798	0.18889

The non-adiabaticity never approaches 2, so the frozen matrix is elliptic at every instant and predicts zero growth in every row, while μ_F approaches $m\Omega_0/2$ in the weak-modulation regime, stays close to it over the range examined, and the last column shows the finite-time rate at $\Omega_0 T = 400$ reproducing it. The geometric speed fails in the same way, since $v_{FS} \propto \eta$ stays small throughout. Replacing $b(t)$ by its modulus suppresses the growth by a factor of forty at $m = 0.02$, which is why the signed coefficient is kept in Eq. (S4). What fails here is locality: parametric resonance is a property of the monodromy over a full modulation period, assembled from coherent phase accumulation across many periods, and neither of the instantaneous diagnostics considered here, the frozen spectrum nor the local geometric speed, encodes it.

S6. Mixed states — bounded distinguishability against divergent variance

For pure states the geometric speed is the energy uncertainty and nothing more. The two quantities separate away from purity, and for the driven mode of Eq. (S3) in a thermal state of mean occupation $\bar{n}$, both are available in closed form. Writing $W \equiv 1 + 2\bar{n} + 2\bar{n}^2$,

$$v = \kappa_s\sqrt{(1 - 1/2W)},\ \Delta\hat{H} = \hbar\kappa_s\sqrt{(W/2)}, \quad \text{(S6)}$$

so that $\hbar v/\Delta\hat{H} = \sqrt{(2W - 1)/W}$. These are a special case of the general Gaussian expressions of Refs. [18–20], quoted in this form because the qualitative separation is then immediate. The two coincide only at $\bar{n} = 0$. Here $\Delta\hat{H}$ grows without bound, approaching $\hbar\kappa_s\bar{n}$ at large occupation, whereas $\hbar v$ is bounded, rising monotonically from $\hbar\kappa_s/\sqrt{2}$ at $\bar{n} = 0$ to $\hbar\kappa_s$ as $\bar{n} \to \infty$, a total variation of $\sqrt{2}$ across the entire temperature range (Fig. S3).

The geometric speed therefore tracks the component of the energy spread quantified by the quantum Fisher information rather than by the total variance — the component that actually generates distinguishability — while the incoherent thermal contribution inflates $\Delta\hat{H}$ without producing any corresponding distinguishability rate. The extreme case is a stationary thermal state of an undriven mode, which has arbitrarily large $\Delta\hat{H}$ and vanishing geometric speed identically.

The speed of a mixed state follows from $v = \tfrac{1}{2}\sqrt{F_Q}$ with $F_Q = (2/\hbar^2)\,\Sigma_{ij}\,(p_i - p_j)^2/(p_i + p_j)\,|\,\langle i|\hat{H}_\perp|j\rangle\,|^2$ in the eigenbasis of the density matrix, whereas $\Delta\hat{H} = \sqrt{(\,\langle\hat{H}_\perp{}^2\rangle\, - \,\langle\hat{H}_\perp\rangle\,^2)}$. Evaluating both for a thermal state under $\hat{H}_\perp$ of Eq. (S3) yields Eq. (S6). The evaluation was performed numerically in a 900-level Fock basis for $\bar{n}$ = 0, 0.1, 0.5, 1, 3, 10 and 30; the maximum absolute deviation from the closed forms is below 10^{-15} for $\bar{n} \le 10$, with a residual of order 10^{-9} at $\bar{n} = 30$ arising from the finite Fock-space truncation.

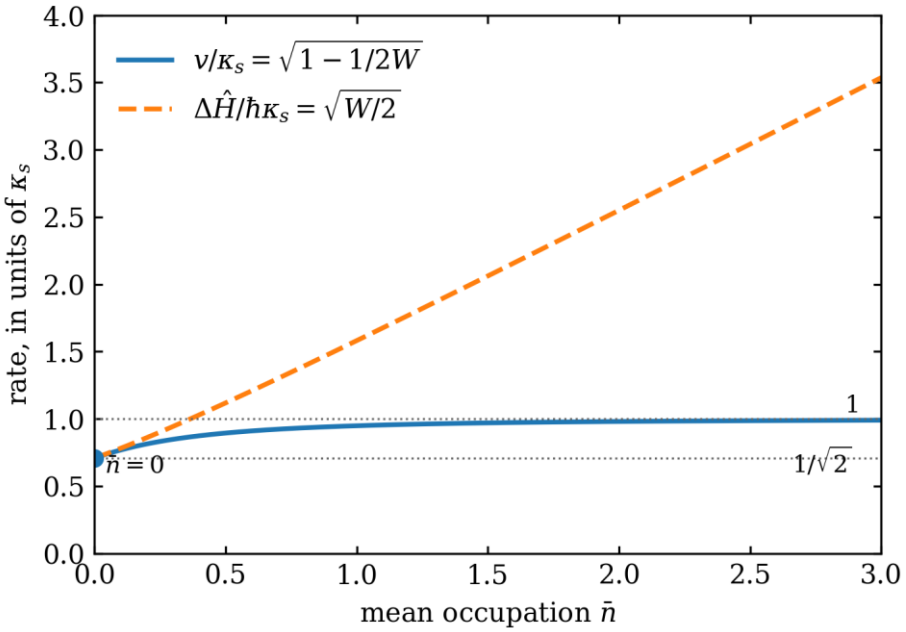


FIG. S3. Geometric speed and energy uncertainty for a driven mode in a thermal state, both in units of $\hbar\kappa_s$ with $\kappa_s = \eta\Omega/2$. Solid: the mixed-state speed v of Eq. (S6), bounded between $\hbar\kappa_s/\sqrt{2}$ and $\hbar\kappa_s$. Dashed: the energy uncertainty of the same state, which grows linearly in $\bar{n}$. The two coincide only at $\bar{n} = 0$, marked by the filled circle, which is the pure-vacuum case; the widening gap shows that the growing thermal contribution to the variance does not produce a corresponding growth of the distinguishability rate.